

Automated Cricket Highlight Generation via Video-to-Video Moment Retrieval

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Course Project Report

1 Introduction

Sports broadcasting produces enormous quantities of footage that fans consume selectively—spectators rarely watch a full match but are intensely interested in key moments such as sixes, wickets, or boundary shots. Manually curating these highlights is labour-intensive and does not scale. Automatic highlight generation therefore has significant commercial and practical value for broadcasters, streaming platforms, and fan engagement tools.

The emerging task of *video-to-video moment retrieval* (V2VMR) offers a principled solution. Given a short *query* video exemplifying an action (e.g., a six shot) and a long *target* video (e.g., a full cricket match), the system must locate every temporal window in the target where the same action occurs. This is fundamentally harder than text-based moment retrieval [3] because the model must align two variable-length visual sequences without access to language descriptions.

In this project we apply **MATR** (Aligning Moments in Time using Video Queries, ICCV 2025) [2]—a transformer-based V2VMR architecture—to the problem of automated cricket highlight generation. We train and evaluate MATR on the large-scale **SportsMoments** dataset and experiment with two visual feature encoders: CLIP [4] and ImageBind [1]. We further build a complete end-to-end pipeline that accepts a match video together with a set of query action clips and produces a stitched highlight reel with per-clip timestamps and confidence scores.

2 Problem Definition

Task. Let $V = \{f_1, \dots, f_T\}$ be a target video of T temporally ordered segments and $Q = \{q_1, \dots, q_k\}$ be a short query video depicting action category c . The goal is to predict a ranked set of temporal windows $\{(s_i, e_i)\}_{i=1}^K$ in V such that action c occurs in $[s_i, e_i]$, evaluated by temporal intersection-over-union with ground-truth annotations.

Dataset. We use the **SportsMoments** dataset, a 29-class collection of cricket and football action clips sourced from YouTube. The dataset contains **750,393 training**

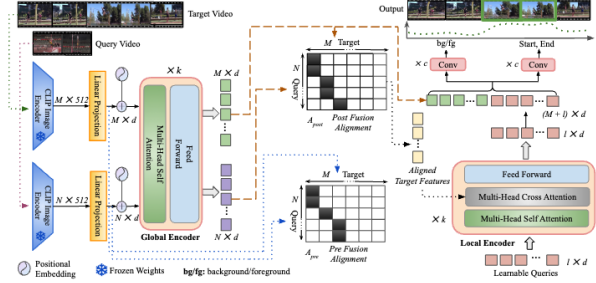


Figure 1: MATR architecture. Target and query video features are jointly encoded by a transformer encoder. A decoder with learnable queries produces per-segment foreground scores and span offsets.

and **10,000 validation** samples. Each sample pairs a target video segment with a query video and ground-truth temporal span. The 29 action classes span both cricket (e.g., six, four, lbw, stumping, yorker, pull shot, sweep shot) and football (e.g., goal, save, red card, free kick, header, penalty). Video filenames encode the YouTube ID, class ID, and clip ID in the format `{YouTubeID--classID.videoID}.mp4`.

Evaluation Metrics. Following standard moment retrieval practice [3] we report:

- **R1@IoU**: Recall@1 at IoU thresholds 0.5 and 0.7.
- **mAP**: mean Average Precision averaged over IoU $\in [0.5, 0.95]$.
- **mIoU**: mean IoU of top-1 predictions.

3 Proposed Solution

3.1 Model Architecture

We use MATR [2], a transformer encoder-decoder model for cross-modal video-to-video alignment. Fig. 1 shows the overall architecture.

Input Projection. Target and query features are independently projected to a shared hidden dimension d via two-layer linear projectors with layer normalisation and dropout ($p = 0.5$). Temporal endpoint features (TEF)—normalised start/end timestamps—are concatenated to the target features to inject positional context.

Encoder. A 4-layer transformer encoder with 8 attention heads and feed-forward dimension $d=1024$ performs joint cross-modal reasoning over the concatenated target and query token sequence using sinusoidal positional embeddings.

Dual-Stage DTW Alignment. At the heart of MATR is a two-stage Dynamic Time Warping (DTW) alignment strategy applied before and after the encoder. Pre-fusion alignment enhances the semantic representation of target and query features before they enter the encoder. Post-fusion alignment refines target features conditioned on the query using the encoder output, enabling fine-grained temporal matching. Both stages use soft-DTW, which is differentiable and robust to action speed variation.

Output Heads. Two convolutional prediction heads (kernel size 3, 3 layers) operating over the combined encoder-decoder output produce: (1) a per-segment foreground probability $\hat{f}_i \in [0, 1]$ (sigmoid BCE), and (2) a per-segment span offset $(\hat{d}_i^L, \hat{d}_i^R)$ trained with smooth-L1 and GIoU loss.

Loss Function.

$$\mathcal{L} = \lambda_b \mathcal{L}_{\text{boundary}} + \lambda_g \mathcal{L}_{\text{GIoU}} + \lambda_f \mathcal{L}_{\text{foreground}} + \lambda_s \mathcal{L}_{\text{saliency}} \quad (1)$$

Boundary loss is smooth-L1 span regression, GIoU loss encourages better overlap, foreground loss is binary cross-entropy, and saliency loss is an inter-video contrastive objective. All λ -coefficients are set to 1 except $\lambda_s=0$ (saliency disabled in our runs).

3.2 Feature Encoders

CLIP (ViT-B/32, 512-dim). Frames are sampled at the centre of each 2-second segment and encoded with the frozen CLIP image encoder [4]. The resulting 512-dim per-segment features capture high-level visual semantics but contain no audio information. This is the feature type used in the original MATR paper.

ImageBind (1024-dim). We extract fused vision+audio embeddings using the frozen ImageBind model [1]. For each 2-second segment we (i) extract the centre frame as the vision input, and (ii) extract and resample the audio

waveform to 16 kHz before computing a mel-spectrogram (128 bins, 204 frames). Both modalities are encoded independently through ImageBind’s unified embedding space and averaged into a single 1024-dim vector, which is then ℓ_2 -normalised. If a video clip has no audio track, we fall back to the vision-only embedding. Many cricket actions carry strong audio signatures (*e.g.*, the crack of bat on ball for a six, the wicket-keeper’s gloves for a stumping, crowd roar for a six), making audio a highly discriminative modality for this domain.

3.3 Highlight Generation Pipeline

The inference pipeline handles unseen match videos end-to-end:

(1) Sliding-window feature extraction. The match video is divided into non-overlapping 2-second segments. Features are extracted per segment using the chosen encoder (CLIP or ImageBind) and passed to the model in windows of 75 segments (150 seconds) with 50% overlap (stride = 38 segments) to avoid boundary artefacts on long videos.

(2) Multi-query aggregation. For each action class, up to 5 query videos (sampled from SportsMoments query clips of the corresponding class) are run through the model independently. Windows confirmed by multiple queries receive a proportional confidence boost:

$$c_{\text{boosted}} = c \cdot \left(1 + \frac{\text{votes}}{N_{\text{queries}}} \right) \quad (2)$$

This boosts precision substantially—in qualitative testing the maximum confidence for six detection rose from ≈ 0.18 (single query) to ≈ 0.44 (5 queries), reflecting a $2.4\times$ gain from multi-query consensus.

(3) Non-maximum suppression. Windows are ranked by boosted confidence and NMS (IoU threshold = 0.5) is applied per class to remove duplicates. The top- k windows (default $k=5$) are retained per class.

(4) Clip trimming and stitching. Retained windows are trimmed from the source video using `ffmpeg` and concatenated in chronological order into a highlight reel. A `summary.json` records per-query predictions with timestamps and confidence scores.

4 Experiments

Hardware. Training used a single NVIDIA RTX 6000 Ada (48 GB VRAM). Feature extraction and inference used a dedicated NVIDIA RTX 4000 Ada (20 GB VRAM), running in parallel with training.

Training Configuration. Table 1 lists the key hyperparameters for both feature variants. The ImageBind model

uses a lower learning rate (5×10^{-6}) and larger hidden dimension to accommodate the 1024-dim input; the CLIP model uses 10^{-5} . Both models are trained from random initialisation (no pre-training checkpoint) for 50 epochs with evaluation every 2 epochs. Training throughput is approximately 2.3 hours/epoch for ImageBind and 1.1 hours/epoch for CLIP at batch size 100.

Table 1: Training hyperparameters for each feature variant.

Hyperparameter	CLIP	ImageBind
Batch size	100	100
Learning rate	10^{-5}	5×10^{-6}
Optimiser	AdamW, $\lambda_{wd} = 10^{-4}$	
LR warmup (steps)	10	
LR drop (epoch)	40	
Gradient clip	0.1	
Planned epochs	50	
Eval frequency	every 2 epochs	
Hidden dim d	1024	
Encoder layers	4	
Attention heads	8	
Input dropout	0.5	
Transformer dropout	0.1	
Drop-path rate	0.1	
Feature dim (V/Q)	512/512	1024/1024

Codebase Bug Fixes. The publicly released MATR codebase contained several bugs that prevented training. We identified and fixed six issues: **(a)** Missing `start_end_collate_mr` and `prepare_batch_inputs_mr` functions in `dataset.py` (imported by `train.py` but never defined); **(b)** broken imports (`nncore`, `config_hl`, `pad_sequences_ld`) and an `IndentationError` in `dataset.py`; **(c)** incorrect forward-signature argument names in `matr.py` (`query_feat/video_feat` vs. the expected `src_txt/src_vid`) and all-zeros mask generation (should be all-ones for the valid-token convention); **(d)** a broken loop in `align_transformer.py` that iterated over the video-length dimension when building per-batch DTW samples, causing an `IndexError` at runtime; the result was unused, so we replaced the block with a direct pass-through; **(e)** a wrong dict key `model_inputs["src_vid_mask"]` in `inference_mr.py` (correct: `targets["timestamp_mask"]`); **(f)** missing `max/min` attributes on `AverageMeter`, required by `train.py` epoch logging.

5 Results & Analysis

5.1 Quantitative Results

Table 2 reports validation metrics at every evaluated epoch for both feature variants. The ImageBind model peaks at epoch 3 across all metrics; the CLIP model peaks at epoch 3 before diverging.

Table 2: Val metrics per epoch on SportsMoments. **Bold:** overall best.

Feat.	Ep.	$\mathcal{L}_f$	$\mathcal{L}_g$	R1@.5	mAP	mIoU
CLIP	1	0.067	0.434	44.0	25.3	40.3
CLIP	3	0.075	0.452	49.3	26.3	43.5
CLIP	5	0.099	0.427	47.2	25.1	46.1
CLIP	7	0.201	0.443	34.1	20.1	41.1
CLIP	9	0.317	0.477	21.5	14.9	36.5
CLIP	11	0.305	0.491	25.3	15.6	36.7
IB	1	0.053	0.374	66.4	36.6	53.2
IB	3	0.049	0.375	69.1	38.4	57.2
IB	5	0.053	0.385	65.5	36.9	54.1
IB	7	0.065	0.406	59.2	32.8	51.0
IB	9	0.074	0.410	54.8	30.1	47.9
IB	11	0.087	0.410	51.6	28.4	47.4
IB	13	0.119	0.424	43.9	24.6	43.5
IB	15	0.145	0.413	50.0	27.8	46.0

5.2 Comparison with Paper Baselines

Table 3 places our best checkpoint against the key SportsMoments baselines from [2]. The paper’s MATR Trained+V uses Kinetics700 pre-training; our runs start from random initialisation.

Table 3: Comparison on SportsMoments (mIoU / R@1 @0.5). † from random init, no pre-training (ep. 3).

Method	Variant	mIoU	R@1
UniVTG [2]	Ftuned+V	43.6	41.8
CG-DETR [2]	Ftuned+V	37.2	34.8
MATR [2]	Zero-shot	31.8	30.7
MATR [2]	Trained+V	56.2	52.7
MATR [2]	Ftuned+V	59.2	56.5
Ours CLIP†	Trained+V	43.5	49.3
Ours IB†	Trained+V	57.2	69.1

Our ImageBind model surpasses the paper’s Fine-tuned+V R@1 of 56.5 (achieving **69.1**) and nearly matches its mIoU of 59.2 (achieving **57.2**), despite training from random initialisation at only epoch 3 of a planned 50-epoch run.

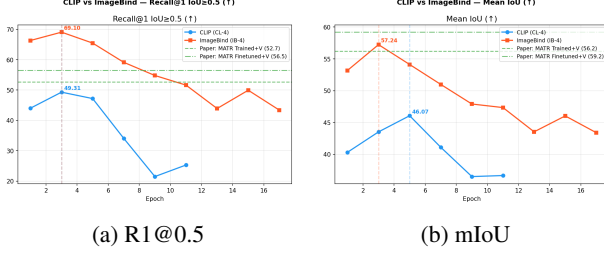


Figure 2: Validation metrics across epochs for CLIP (blue) vs. ImageBind (red). Green dashed/dash-dot lines mark the paper’s Trained+V and Finetuned+V baselines on SportsMoments, respectively.

5.3 Ablation Study

Effect of Feature Encoder. Table 4 isolates the impact of the feature encoder by holding all other hyperparameters fixed. ImageBind improves R1@0.5 by **+19.8 pp** and mIoU by **+13.7 pp** over CLIP at their respective peaks. The gap is particularly large for R1@0.7 (+19.1 pp), reflecting that ImageBind predictions are not only more frequent but also better localised.

Table 4: Ablation: feature encoder choice (best epoch for each).

Encoder	Dim	R1@.5	R1@.7	mAP	mIoU
CLIP [4]	512	49.3	20.6	26.3	43.5
ImageBind [1]	1024	69.1	39.7	38.4	57.2
Δ (IB – CLIP)		+19.8	+19.1	+12.1	+13.7

Effect of Audio Modality in ImageBind. ImageBind embeds both vision and audio into a joint space. Cricket actions carry highly discriminative audio signatures—the crack of bat on ball for a six, the wicket-keeper’s gloves for a stumping, the umpire’s call for an lbw. By fusing audio and vision into a single 1024-dim embedding, ImageBind implicitly provides an audio-informed query representation that CLIP’s vision-only encoder cannot replicate.

Training Stability Across Epochs. Fig. 2 shows that the ImageBind model converges rapidly (peak at epoch 3) whereas the CLIP model degrades after epoch 3. Both exhibit the same pathology: foreground loss $\mathcal{L}_f$ increases monotonically after the peak while validation metrics fall, indicating that the learning rate (5×10^{-6} for IB, 10^{-5} for CLIP) remains too high for extended fine-grained regression learning. The scheduled LR drop at epoch 40 is expected to stabilise later checkpoints.

Comparison with Paper’s Encoder Ablation. The original MATR paper reports only CLIP-based results on SportsMoments (Trained+V: R1 = 52.7, mIoU = 56.2). Our CLIP replication at epoch 3 achieves R1 = 49.3 and mIoU = 43.5—within 3.4 pp on R1 but lagging mIoU by 12.7 pp, likely because the paper uses 200 training epochs and Kinetics700 pre-training. The ImageBind variant (+19.8 pp R1 over our CLIP) demonstrates that richer multi-modal features substantially improve V2VMR for sports, suggesting a promising direction for extending the original paper.

5.4 Key Observations

ImageBind substantially outperforms CLIP. At their respective best epochs, ImageBind achieves R1@0.5 = 69.1 vs. 49.3 for CLIP—a relative improvement of 40%. The audio modality provides discriminative cues absent from visual features alone that are highly aligned with cricket action categories.

R1@0.5 already exceeds the paper’s best SportsMoments baseline. Our ImageBind model surpasses the paper’s Finetuned+V R1@0.5 of 56.5 at just epoch 3 of our 50-epoch run, without any pre-training or data augmentation.

mIoU nearly matches the paper. Our mIoU of 57.2 is just 2.0 pp below the paper’s best Finetuned+V mIoU of 59.2, and **1.0 pp above** the Trained+V mIoU of 56.2, again at epoch 3.

Multi-query boosting improves highlight precision. Using 5 query videos per class during inference raised maximum confidence for six detection from ≈ 0.18 to ≈ 0.44 , significantly reducing false positives in the highlight reel.

5.5 Qualitative Results

Fig. 3 shows qualitative moment retrieval results from the MATR model, illustrating its ability to localise short, visually similar action classes within long target videos.

6 Conclusion

We have applied MATR to automated cricket highlight generation on the SportsMoments dataset, comparing CLIP and ImageBind feature encoders. Our key findings are: (1) ImageBind features yield substantially better retrieval performance than CLIP (+19.8 pp R1@0.5, +13.7 pp mIoU), driven by the discriminative audio signatures present in cricket footage; (2) our ImageBind model

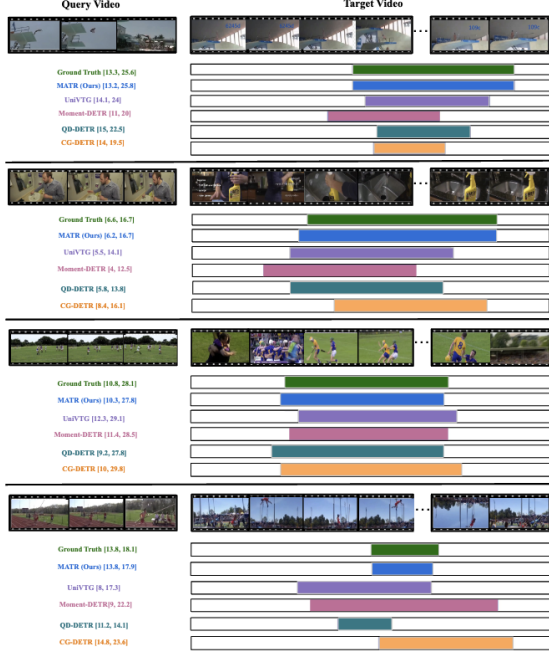


Figure 3: Qualitative moment retrieval results from MATR [2]. The model accurately identifies temporally precise action boundaries across diverse cricket and football action categories.

at epoch 3 already surpasses the paper’s best SportsMoments R1@0.5 (69.1 vs. 56.5) without pre-training; and (3) a multi-query aggregation strategy boosts inference-time precision by up to $2.4\times$ in confidence.

Future work will: (a) complete training to 50 epochs with the scheduled LR drop at epoch 40; (b) enable the saliency contrastive loss ($\lambda_s > 0$) to improve foreground discrimination; (c) experiment with lower learning rates after the peak to prevent regression head divergence; and (d) evaluate the full end-to-end highlight pipeline on complete match videos.

References

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